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Implementation of Low Frequency Band Pass Filter by 3-OTA based Floating Active Inductor

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***Abstract*—Thispaper presents a competitive alternative to conventional passive spiral grounded inductor by realising a 3-OTA CMOS floating active inductor circuit. The realised floating active inductor has been used to implement a low frequency band pass filter design. The designed floating active inductor shows a high frequency operating range with wide tuning capability. The simulation of the proposed grounded active inductor and implementation of band pass filterin low frequency domain are carried out by using Cadence Virtuoso UMC 180 nm technology by spectre simulator.**

Keyword: Floating Active Inductor, BPF, SFG, OTA

I. Introduction

An inductor is one of the basic electronic components, frequently used for designing filters, low-noise amplifiers (LNA), matching circuits, and different kinds of oscillators and so on. To use an inductor in an integrated circuit (IC) introduces a significant drawback in terms of large chip areas. Relative chip areas required for fabricating passive inductors have not been decreased even in the ultramodern CMOS technology, where minimum gate length is in deep sub-micron [1]. Again Q factor of an on-chip inductor is relatively low. A number of research papers address the above issue. The use of active inductor is one of the most successful and widely used approaches to overcome the above mentioned problem [1].

An active inductor is a circuit whose input impedance is the same as that of the passive inductor [2, 3]. It occupies very small relative chip area as compared to the passive one as it consists of MOSFETs and its parasitic capacitors. A floating active inductor is a one port network for which the input impedance is same as that of the spiral passive inductor and the two terminals are always floating as shown in fig 1.

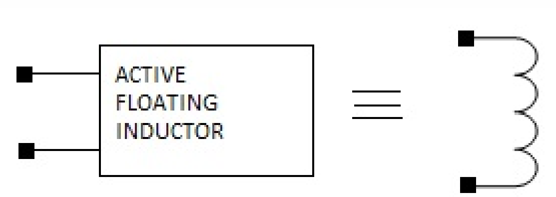

Fig. 1: Floating Active Inductor

To design a floating active inductor, operational transconductance amplifier (OTA) circuit [1, 4] is widely used. The implemented floating active inductor is the modification of the architecture reported in [4, 3, 6] in terms of power consumption as well as tuning range. This paper is organised as follows: section II deals with basics of OTA cell. Section III describes the designed floating active inductor implementation. Section IV deals with testing and simulation results of the designed floating active inductor architecture. Section V deals with the implementation of low frequency BPF design and the paper isconcluded insection VI.

II. Basics of Cmosota Cell

This section deals with the basics of Operational Transconductance Amplifier which is the mail building block to design a floating active inductor. The fig 2 [5, 8] shows the schematic of differential OTA with its circuit symbol.

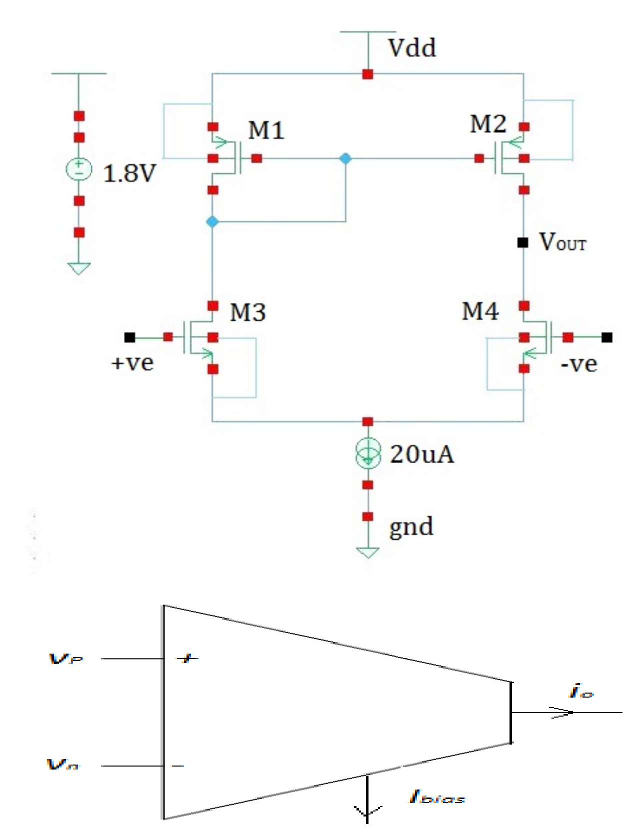

Fig. 2: Schematic of OTA Cell with Circuit Symbol

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The aspect ratios are shown in Table 1.

TABLE 1

Transistors	W/L (um/um)	W/L (um/um) [4]
M1	20/0.9	30/5
M2	17/0.9	25/5
M3-M4	20/0.9	30/5

The OTA uses one current source and 4 MOS transistors where M3-M4 is perfectly matched transistors and M1-M2 are mirrored. All the transistors are operating in saturation [4, 8].

III. 3-OTA BASED FLOATING ACTIVE INDUCTOR IMPLEMENTATION

Figure 3 shows the 3-OTA based floating active inductor implementation [4, 6].

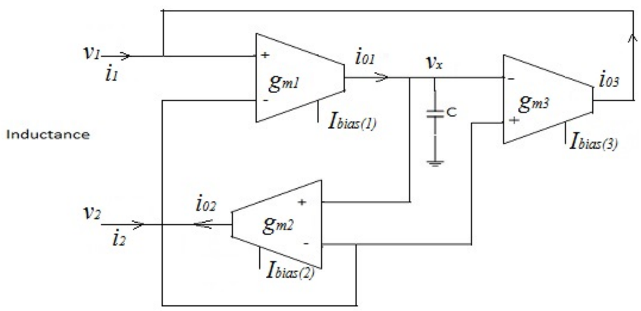

Fig. 3: 3-OTA based Floating Active Inductor Architecture

The output current i_{01} of OTA1 is expressed as

$$i_{01} = g_{m1}(v_1 - v_2)` \tag{1}$$

$$\text{Again, } g_{mn} = \frac{I_{bias(n)}}{2v_t}\ ;\ n = 1, 2, 3 \tag{2}$$

Where v_t is the thermal voltage. Now the voltage across the capacitor id defined as

$$v_x = \frac{i_{01}}{sC} \tag{3}$$

Now,

$$i_{02} = g_{m2}\left(\frac{i_{01}*(v_1 - v_2)}{sC} - v_2\right)$$

$$= \frac{g_{m1}g_{m2}v_1}{sC} - \frac{g_{m1}g_{m2}v_2}{sC} - g_{m2}v_2 \tag{4}$$

As $g_{m1}g_{m2}/\omega C \gg g_{m2}$, ignoring the 3rd term of the above equation, we get

$$i_{02} = \frac{I_{bias(1)}I_{bias(2)}}{4v_t^2 sC} * v_{1-2} \tag{5}$$

$$\text{Again, } i_{02} = -\frac{I_{bias(3)}}{I_{bias(2)}} * i_{03}$$

But, in the circuit design, all the bias currents are same in magnitude. So,

$$i_1 = i_{02} = -i_{03} \tag{6}$$

$$\text{And } i_2 = -i_{02} \tag{7}$$

Again i_1 and i_2 are same in magnitude but opposite in phase. Now the required inductance value between terminal 1 and 2 is observed by calculating impedance

$$Z_{1-2} = \frac{4v_t^2 sC}{I_{bias(1)}I_{bias(2)}} = \frac{sC}{g_m^2} = sL_{equivalent} \tag{8}$$

So, one can control the inductance value by controlling bias current as well as C value.

The fig 4 shows the schematic of 3-OTA based floating active inductor LR circuit.

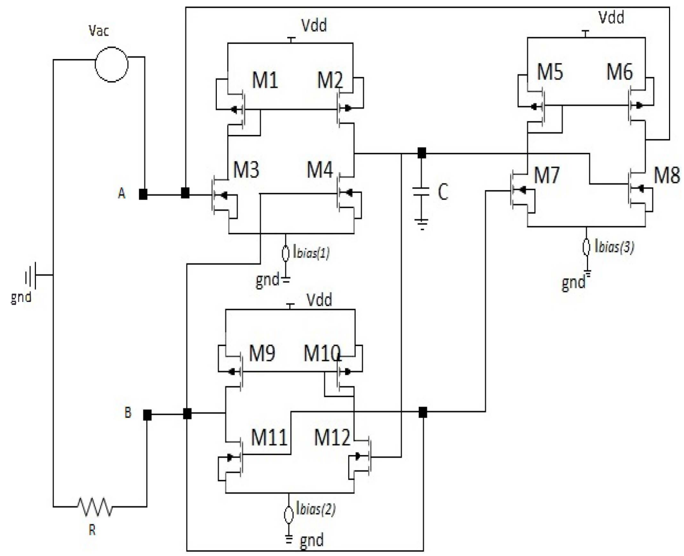

Fig. 4: 3-OTA based Floating Active Inductor Schematic of LR Circuit

IV. TESTING AND SIMULATION RESULT

A 10 μV sine wave of 100 kHz frequency with zero initial phase are applied as an input in the presence of 1.8 V Vdd across node A. After transient analysis, the obtained voltage and current waveforms are given in Fig. 5.

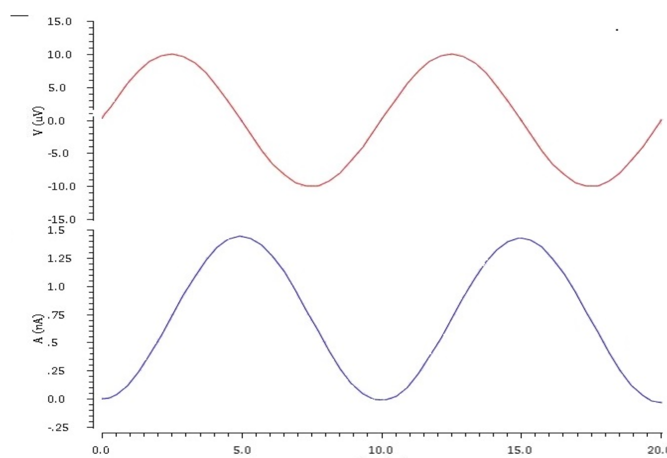

Fig. 5: Voltage and Current Waveforms Across Node A

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From this graph it is clear that the current lags voltage by 90 degree. So, the designed architecture is undoubtedly acts as an active inductor.

With the same input configuration, the SP analysis followed by ZM analysis in between node A and B terminal is performed to obtain impedance vs. frequency plot [3, 2] as shown in Fig. 6.

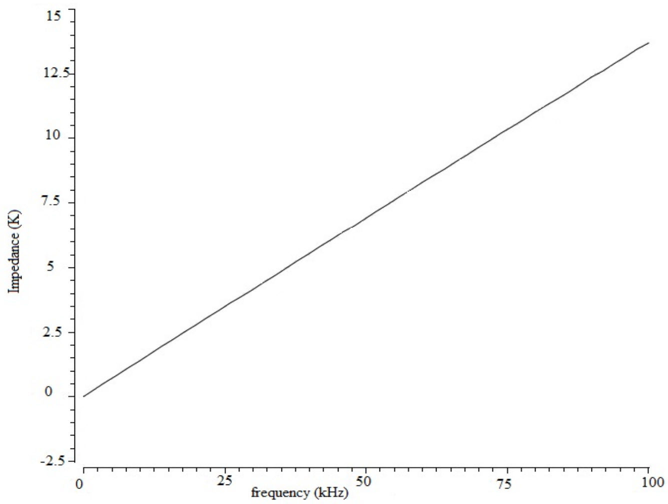

Fig. 6: Impedance vs. Frequency Plot

From this plot it is clear that linearity approaches up to 100 kHz that is the inductive nature is obtained up to 100 kHz. Again from the following equation

$X_L = j2\pi f L$

The obtained practical inductance value is 21.9 mH which is high as reported in [4, 6].

The total power dissipation is 245uW.

V. Low Frequency BPF Design

The Fig. 7 shows the passive implementation of series floating inductor based BPF:

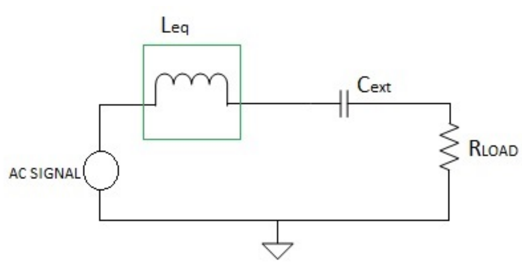

Fig. 7: Passive Floating Inductor based BPF

In this figure R_{LOAD} and C_{ext} are out of floating active inductor chip. $L_{eq}$is replaced by 3-OTA based floating active inductor chip as mentioned in fig4. The aspect ratios are given in table 2. The$L_{eq} = 25mH$, $R_{load} = 600\ Ohms$and $C_{ext} = 100nF$ were set. To obtain 25mH inductance value, the grounded capacitor value of 3-OTA based floating active inductor is selected 11.41 pF.

Table 2: Aspect Ratio of Transistors for Designing 3-OTA Floating Active Inductor based Series RLC BPF

Transistors	W/L Ratio (um/um)
M1-M2	25.5/0.9
M3-M4	20/0.9
M5-M6	26/0.9
M7-M8	23.5/0.9
M9-M10	27.5/0.9
M11-M12	22.5/0.9

The complete block diagram is given in Fig. 8.

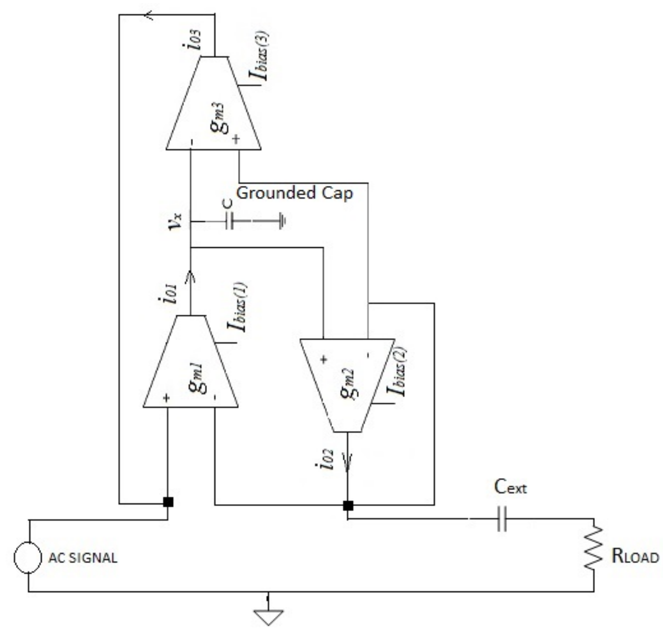

Fig. 8: 3-OTA Floating Active Inductor based RLC BPF

To obtain the BPF characteristics, the ac analysis across load resistance has been performed to obtain gain vs frequency plot as shown in Fig. 9.

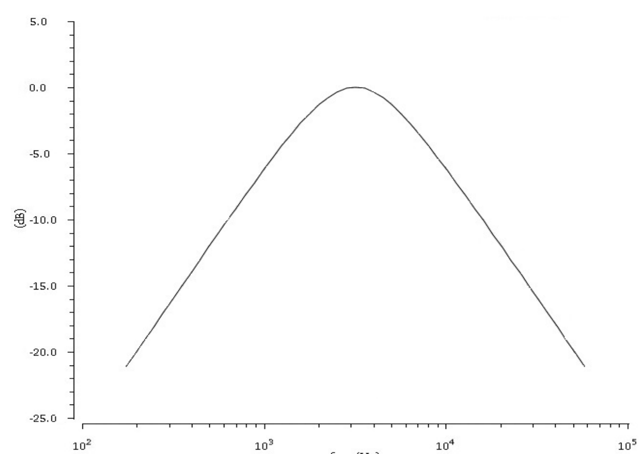

Fig. 9: Gain *vs*. Frequency Plot of Designed BPF

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From this plot, the lower 3-dB cut-off frequency is 1.53633 kHz and upper 3-dB cut-off frequency is 6.63059 kHz with practical resonant frequency 3.1859 kHz. The theoretical resonant frequency is 3.18309 kHz obtained.

VI. CONCLUSION

This paper presents a low transistor count, high value and easily tuned modified floating active inductor architecture with high frequency operating range. The designed inductor is further used to design a low frequency BPF by using QFG MOS architecture.

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